RESEARCH SERVICES DIVISION

FULL-BLOWN RESEARCH PROPOSAL

Proposed Title of Research: A Localized Python-Based Monte Carlo Simulation Model for Assessing Copra Farmer Income Risk and Variability under Price and Yield Uncertainty in Camarines Norte.			
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Classification				Mode of Implementation	
Research		Development			
	Basic	/	Pilot Testing	/	Individual
/	Applied		Tech Promotion/ Commercialization		Interdisciplinary

UCN Research Agenda			
/	Food Security and Poverty Reduction		Pollution Control
	Education and Capability Building		Competitiveness
	Community based/participatory		Natural Resources Management
/	Technology Development	/	Information, Communication Technology
	Health and Nutrition		DRRM / Climate Change Adaptation

Rationale/Significance

State the research problem and significance of the proposed study to the current needs of the country. The proposal should clearly discuss the specific needs/ problems to be addressed the proponent should have a clear understanding of the needs/ problems in the research area of interest. Benefits of research in the institution shall be discussed.

Copra production is one of the major agricultural activities in the Philippines and a major source of income of farming communities especially in provinces where coconut farming is prevalent. **Copra production is one of the main sources of income for many farmers in Camarines Norte.** But because the price of copra, the coconut yields and the cost of production can change during the production cycle, the income from copra is uncertain. The value of copra sold by farmers may be directly affected by changes in market prices. Variations in yields can be related to variations in farm productivity, weather, seasonal factors and other production related

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conditions. Such uncertainties and variations could make it difficult for farmers to estimate their income and prepare for adverse financial conditions.

Income of copra farmers can be regarded as a complex system, in which several interrelated factors with different levels of predictability are involved. For example, if a farmer's yield is low, it will result in lower income regardless of prevailing prices. On the other hand, changes in copra prices can affect gross income substantially if production volumes remain the same. Similarly, increases in production costs can offset total income if copra yield and prices are favorable. Thus, it can be observed that a combination of factors can lead to different outcomes. While using the mean values of prices, yields, and production costs, one can estimate the expected income level with a certain probability. However, the methods described above are limited in the sense that they do not consider different scenarios that may occur with various levels of likelihood. Therefore, it is essential to choose an appropriate tool for the task that will allow considering uncertainties and calculating the probabilities of different incomes.

Monte Carlo Simulation (MCS) is an accurate method to evaluate the impact of random prices and yields on the income of copra farmers by considering the expenses in the calculation of net income. It is possible to create numerous combinations of copra prices and yields using MCS, as it employs simulation procedures that require a probability distribution function to generate values. Additionally, the production costs are important and should be deducted from the gross income to illustrate the net income appropriately. The described MCS methodology could be considered beneficial for calculating the expected value of net income, standard deviation, range, and the probability of different levels of income below the mean. Compared to a specific value, the range presented by MCS results can provide copra farmers with more information about their potential net profit concerning prices, yields, and expenses.

Therefore, this study will propose the construction of the **Localized Python-Based Monte Carlo Simulation Model for Assessing Copra Farmer Income Risk and Variability under Price and Yield Uncertainty in Camarines Norte.** It is expected that the abovementioned simulation model will utilize copra yield and price as input variables with production costs considered in the computation of the net income. The model will also help determine the potential impact of copra price and yield changes on the farmers' net-income, considering production costs. Several simulation experiments can be designed to test how different prices and yields would affect the income. The results of the simulation will provide empirical evidence needed by the farmers and stakeholders to understand possible risks and address them accordingly. Aside from that, this study will also present the use of computer programming, statistical analysis, mathematical, and simulation techniques to solve agricultural problems in the real world, providing an opportunity for further research and development of technological tools that can aid in agricultural production decision-making.

Objectives

General Objective

The study aims to develop and evaluate a **localized Python-based Monte Carlo Simulation model for assessing copra farmer income risk and variability under price and yield uncertainty in Camarines Norte, with production costs incorporated in the calculation of net income**

Specifically, the study aims to:

1. **Identify and characterize the key factors affecting copra farmer income in Camarines Norte,** particularly copra price and production yield, including their historical variability and relevant probability distributions, while accounting for production costs in the calculation of net income.
2. **Collect and analyze relevant historical or observed data on copra prices, coconut yield, and production costs to establish the appropriate input parameters for the simulation model.**

3. **Develop a Python-based Monte Carlo Simulation model** representing potential net-income outcomes of copra farmers under uncertain copra prices and yields, with production costs deducted from gross revenue.
4. **Verify and validate the simulation model** by checking its implementation and comparing simulated results with available historical or observed income, price, and yield data to determine whether the model produces reasonable and representative outcomes.
5. **Estimate the variability and distribution** of potential copra farmer net income using simulation-generated measures such as:
    - **expected net income**
    - **minimum and maximum simulated net income**
    - **standard deviation**
    - **income percentiles**
    - **probability of low-income outcomes**
    - **confidence intervals**
6. **Determine the individual and combined effects of price and yield uncertainty** on copra farmer net income through controlled simulation scenarios while accounting for production costs.
7. **Develop and simulate alternative price and yield scenarios**, including favorable, typical, and unfavorable conditions, to examine their potential effects on farmer net income.
8. **Compare the simulated net-income outcomes** across different scenarios in terms of income variability, probability of unfavorable income outcomes, and overall financial risk.
9. **Identify the conditions that contribute** most significantly to copra farmer income risk based on the results of the simulation experiments.
10. **Evaluate the usefulness of the developed simulation model** as a decision-support tool for understanding copra income risk and supporting agricultural planning and risk-management decisions in Camarines Norte.

Methodology
(Brief description of how the study will be implemented)

The methodology includes the methods, procedure and techniques used to collect and analyze information. It should include the research design, participants and process of selecting sample

respondents, validation and reliability of the instrument/s to be used, methods and data collection, data analysis and procedure and statement how ethical issues will be addressed.

The study will employ an applied quantitative modeling-and-simulation research design using a Monte Carlo Simulation implemented in Python to assess copra farmer income risk and variability under price and yield uncertainty in Camarines Norte. Monte Carlo simulation is appropriate for quantitative risk analysis because it represents uncertain input variables through probability distributions and repeatedly simulates possible outcomes to estimate the range and likelihood of potential results (Bandyopadhyay, 2022). The study will begin by defining the system and its possible boundaries, identifying the essential variables, collecting and statistically representing the required input data, developing and implementing the model, and then verifying, validating, and analyzing its outputs. The system to be modeled is the income-generating activity of copra farmers in Camarines Norte. **Copra yield and copra selling price will serve as the two primary uncertain input variables of the Monte Carlo model, while production costs will be incorporated as an additional stochastic input in calculating estimated net income as the principal output.** Consistent with the principle of abstraction in modeling, only variables necessary to answer the study objectives will be included instead of reproducing every aspect of coconut farming, which could result in unnecessary model complexity.

Participants of this study will be the selected copra farmers in the major coconut producing areas of Camarines Norte, together with also the selected informants who have relevant or same knowledge of local coconut production or copra trading (local agricultural, farmer-organization). These are selected using an appropriate sampling procedure based on the identified target population, consistent with the recommendation that purposive sampling be used when a researcher deliberately selects information-rich respondents who possess direct, first-hand knowledge relevant to the specific variables under study (Memon et al., 2025). Primary data will be collected through a structured questionnaire and interview/data-recording form covering production or yield, selling prices, production costs, and other information necessary to establish the parameters of the simulation. Secondary data will be obtained from relevant official and local sources, particularly the Philippine Coconut Authority (PCA) and Philippine Statistics Authority (PSA), consistent with the current proposal. Before field administration, the instrument will undergo expert content validation and pretesting to ensure clarity, relevance, completeness, and suitability for the variables required by the model. For the reliability of the data collecting process will be supported through standardized administration,

consistent units of all the measurements and also checking of the responses for completeness and with consistencies, and lastly also with comparison with available records if possible.

After the data-collection process, the collected data will be cleaned, organized, and statistically characterized before being used as model inputs. Consistent with the proposal, yield will initially be modeled using a triangular distribution based on minimum, most-likely, and maximum values obtained from the collected field data. Copra price will be modeled using a log-normal distribution based on observed local price data. Production costs will likewise be statistically analyzed and represented using an appropriate probability distribution based on observed field data. The selected probability distributions will be assessed using descriptive statistics and appropriate goodness-of-fit approaches before being incorporated into the Monte Carlo simulation.

The core income equation to be implemented in the model is:

Net Income = (Yield × Selling Price) − Total Production Costs

where **Yield** represents the quantity of copra produced in kilograms per farmer per production cycle, **Selling Price** represents the copra selling price in Philippine pesos per kilogram, and **Total Production Costs** represent the production expenses incurred during the corresponding production cycle. The product of yield and selling price represents the farmer's gross revenue, while subtracting total production costs provides the estimated net income. **This formulation allows the model to examine how uncertainty in yield and selling price, together with stochastic variation in production costs, affects possible farmer net-income outcomes.**

The conceptual model will then be translated into an algorithm, pseudocode, flowchart, and Python program, following the programming sequence specified in the course material. **For each Monte Carlo iteration, the program will generate random values for yield, selling price, and production costs from their respective probability distributions and use these values to calculate a possible farmer net-income outcome.** The simulation will initially be run for 10,000 iterations as a starting point. **Simulation adequacy will then be assessed through convergence and stability checks by monitoring the running mean and standard error of simulated net income at predetermined intervals.** If the results continue to fluctuate beyond the predefined stability criterion, additional iterations will be performed until sufficient stability is achieved.

The model will undergo verification and validation to assess if the computerized model correctly implements the conceptual and mathematical model and if the resulting simulation is adequately representative of the real world system for its purpose. Verification and validation are recognized as essential components in establishing the acceptability and credibility of simulation models (Biazen et al., 2025). Verification will be conducted through code checking, test cases, and step-by-step execution to ensure that the python program correctly implements the conceptual model with the mathematical equations, also the probability distributions and the simulation procedures. It will then be validated by comparing simulated results with available historical or observed local data, using a Kolmogorov–Smirnov goodness-of-fit test applied with care to avoid common misapplications of the one-sample test in practice (On Misuses of the Kolmogorov–Smirnov Test for One-Sample Goodness-of-Fit, 2024), with revisions made when necessary. After validation, Monte Carlo simulations will examine price uncertainty, yield uncertainty, their combined effects, and favorable, typical, and unfavorable scenarios. Outputs will be analyzed using expected income, minimum and maximum income, standard deviation, percentiles, probability of low-income outcomes, confidence intervals, and scenario comparisons. Simulation adequacy will be assessed through the stability of results across additional iterations or replications. Finally, the usefulness of the developed simulation model as a decision support tool will be evaluated based on its ability to present clear and understandable and locally relevant information on copra farmer income variability and risk that can support agricultural planning and risk management decisions in Camarines Norte.

Required permissions will be secured before data collection. Participation will be voluntary, and respondents farm, price, cost, and income information will remain confidential and reported only in aggregate.

Overall, the study will follow the modeling-and-simulation process of defining the system, collecting and representing inputs, developing and implementing the model, verifying and validating it, and analyzing simulation outputs. The resulting Python model will estimate copra farmer income variability and risk under uncertain price and yield conditions in Camarines Norte.

RESEARCH SERVICES DIVISION

Activities	Duration (Timeline)																	
	W1	W2	W3	W4	W5	W6	W7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16	W 17	W 18
1. Finalization and submission of proposal	/	/																
2. Approval, permits, and coordination		/	/															
3. Validation and pretesting of instruments			/	/														
4. Preparation of data-gathering materials				/	/													
5. Farmer interviews and data collection					/	/	/	/										
6 Secondary data collection					/	/	/	/										
7 Data encoding, cleaning, and statistical preparation								/	/	/								
8 Model formulation and Python development										/	/	/	/	/				
9. Model verification and validation														/	/	/		
10. Monte Carlo simulation, scenario analysis, and interpretation															/	/		
11. Final writing, presentation, and editing																/	/	/
12. Printing, submission, and terminal report																	/	/

Expected Output
(Indicate the specific products, processes or services which the study is expected to produce and how these
can be used; quantify when possible)

The study will produce a localized Python-based Monte Carlo Simulation model that simulates the potential net income of the copra farmers considering various copra prices and coconut yields in the province of Camarines Norte and netting off the production costs in order to come up with the final net proceeds. The proposed model will simulate potential income outcomes, probability distributions, the expected net income, the lowest and highest simulated income, standard deviation, various income percentiles, probabilities of low-income occurrences, confidence intervals, and other relevant statistics that characterize the income risks and volatility of the copra farmers based on the relevant historical price, yield, and production cost data that will be gathered and utilized in this study. The Python-based Monte Carlo simulation model will be subjected to various simulation iterations to generate a range of projected net-income outcomes to evaluate the stability of the results of the simulation.

The proposed crop income model will be used by copra farmers, agricultural planners, researchers, and local government units (LGUs) as a decision-making tool that will help the targeted end-users understand the risks associated primarily with price and yield fluctuations while factoring in the production costs to net proceeds. The crop income model will help the targeted end-users evaluate possible favorable, average, and adverse price-yield scenarios, identify the likelihood of low-income occurrences, and assess the impact of changes in prices and yields on the income of the copra farmers. The study will also produce a Python-based Monte Carlo simulation code that can be updated and modified by the targeted end-users to leverage new data to evaluate continuously the income risks and volatility that will be beneficial in studying the income risks of the copra farmers in Camarines Norte.

Implementing Schedule	Month	Year	Duration in Months
Planned Start Date	August	2026	
Planned Completion Date	December	2026	5

Budgetary Requirement

Particulars	Source of Funds				
	PS	MOE	INC	CO	TOTAL
Traveling Expenses (Meeting & Survey)					₱1000
Printing and Photocopying Expenses					₱500

Internet & Data Expenses (1 Month)					₱1000
Research Instrument Validation and Pretesting					₱1000
Software (IDE)					₱0
Miscellaneous (Unexpected Expenses)					₱500
Total materials, supplies and equipment					
B. Other Operating Expenses					
Total Budgetary requirements					₱4000

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adopted from DOST Forms 1A and 1B Capsule RD Proposal with modification